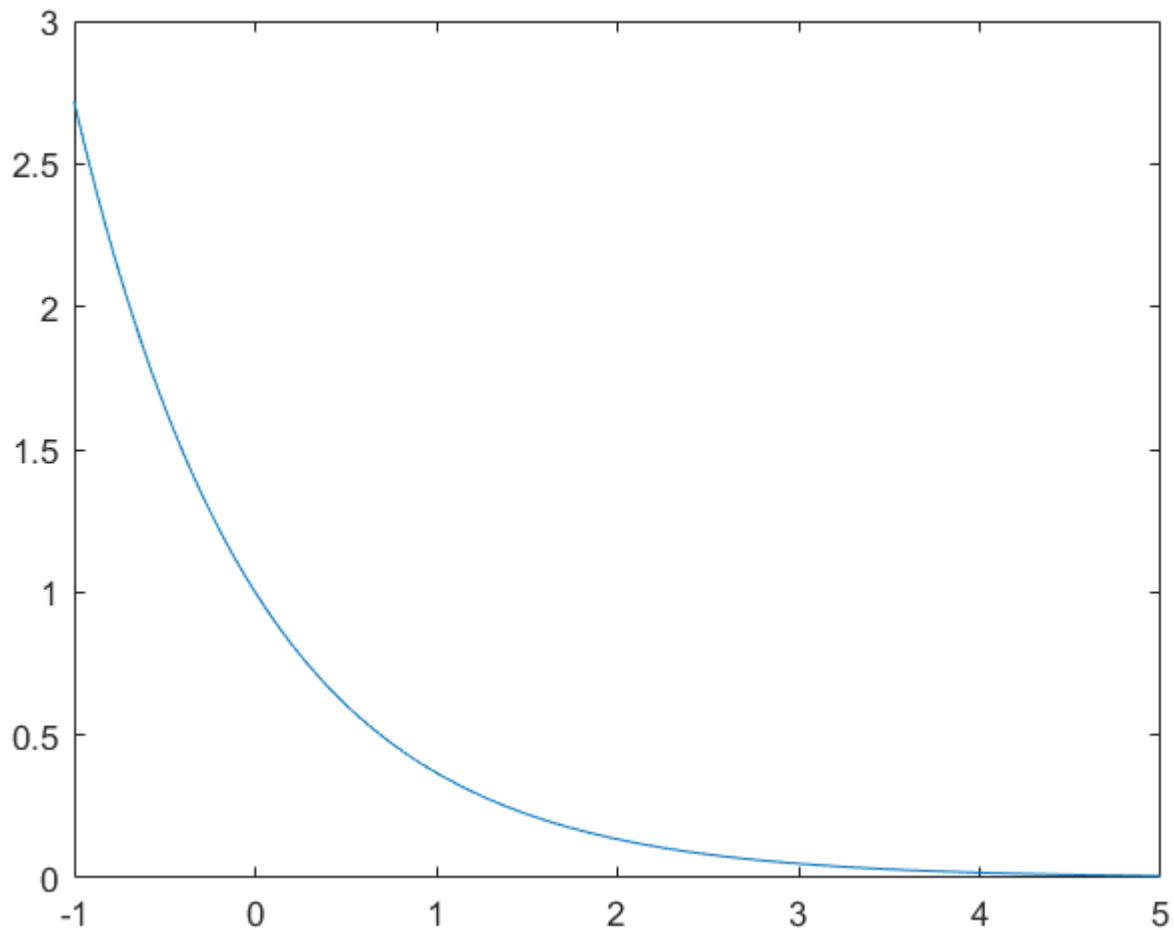


Lab 13 - Multistep and Predictor Corrector Methods

Name: Lilith Janevska

Problem 1

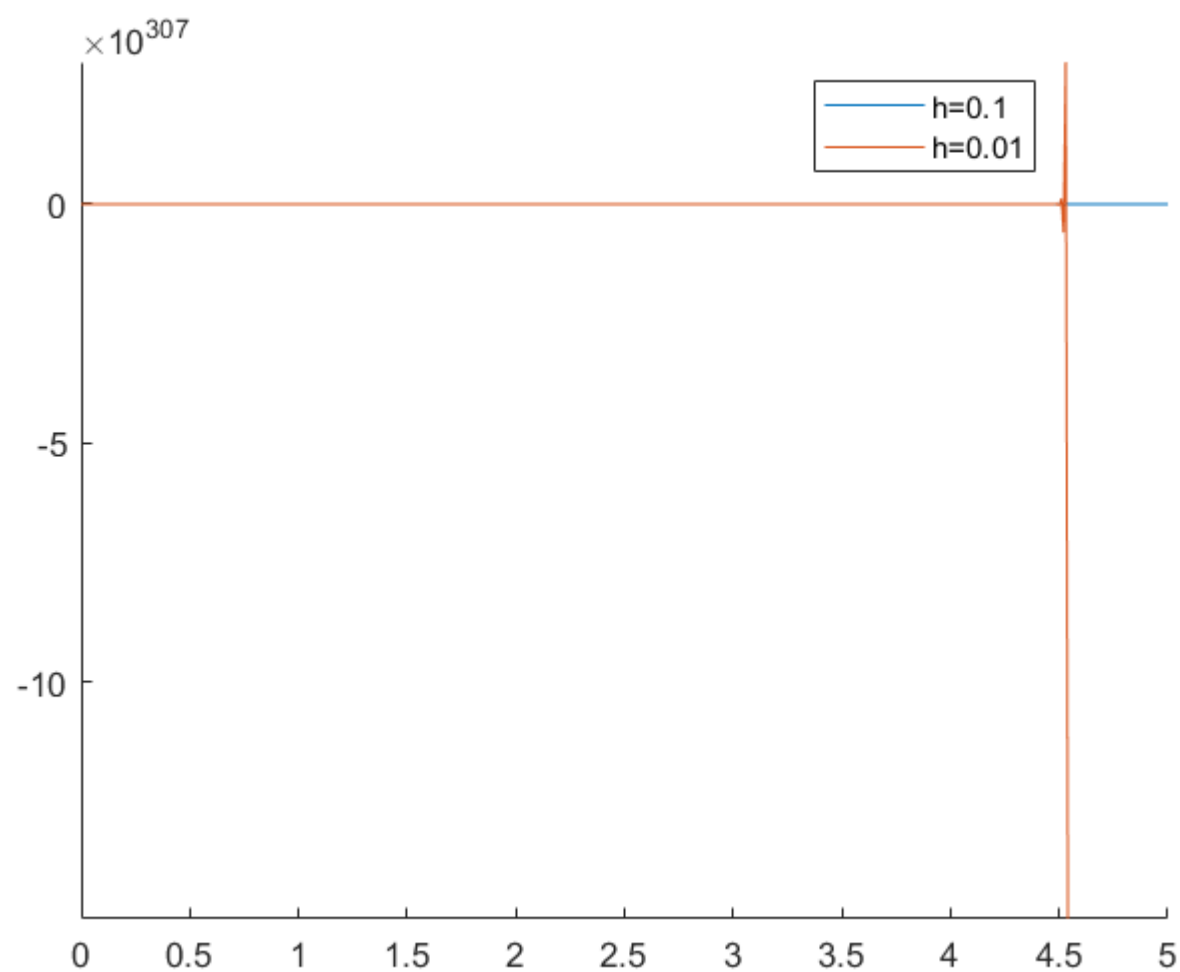
a)



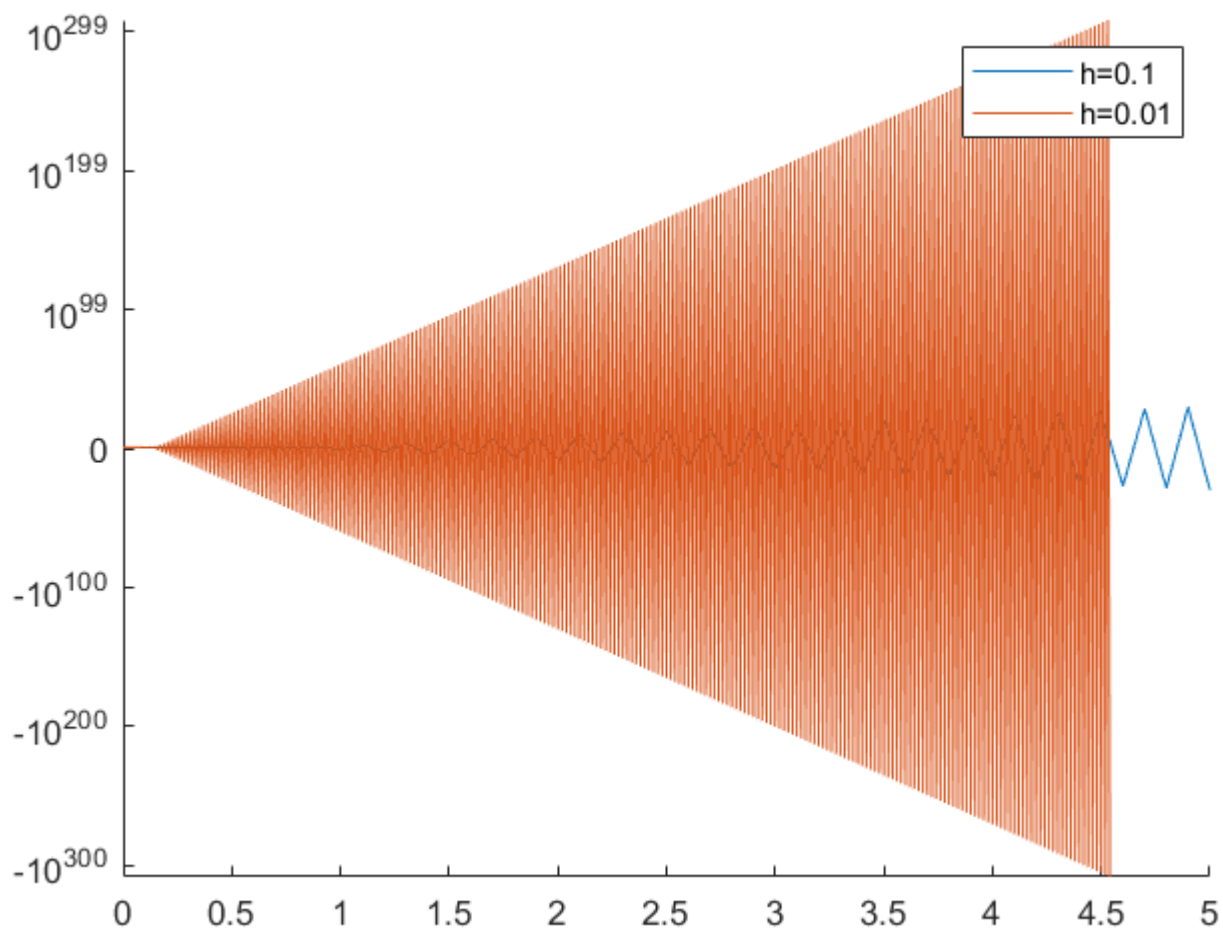
b)

```
function [result] = TwoStepMethod(f,y0,y1,t0,tn,h)
    yn = [y0; y1];
    i = t0;
    while i < tn
        v1 = f(i,yn(end,1));
        v2 = f(i-h,yn(end-1,1));
        yn = [yn; -4*yn(end,1) + 5*yn(end-1,1) + h*(4*v1 + 2*v2)];
        i = i + h;
    end
    result = yn;
end
```

Output:



As a log plot:

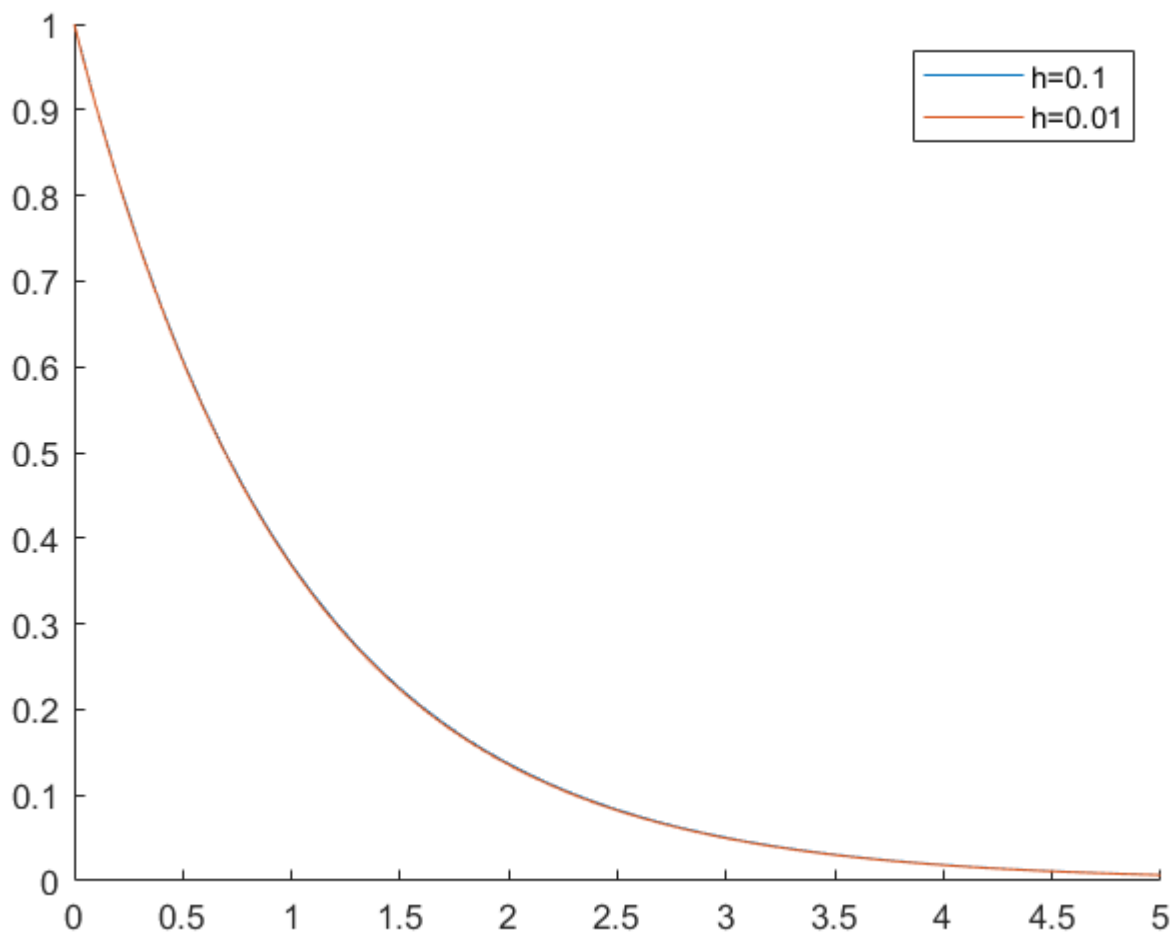


The log plot makes it much more easy to see that there is this oscillatory behaviour present, which should not be occurring! The reason why is because this method and differential equation do not guarantee convergence with every h value. It likely only works for a given set of h values, or perhaps none!

c)

```
function [result] = ABMethod(f,y0,y1,t0,tn,h)
    yn = [y0; y1];
    i = t0;
    while i < tn
        v1 = f(i,yn(end,1));
        v2 = f(i-h,yn(end-1,1));
        yn = [yn; yn(end,1) + h*((3/2)*v1 + (-1/2)*v2)];
        i = i + h;
    end
    result = yn;
end
```

Output:

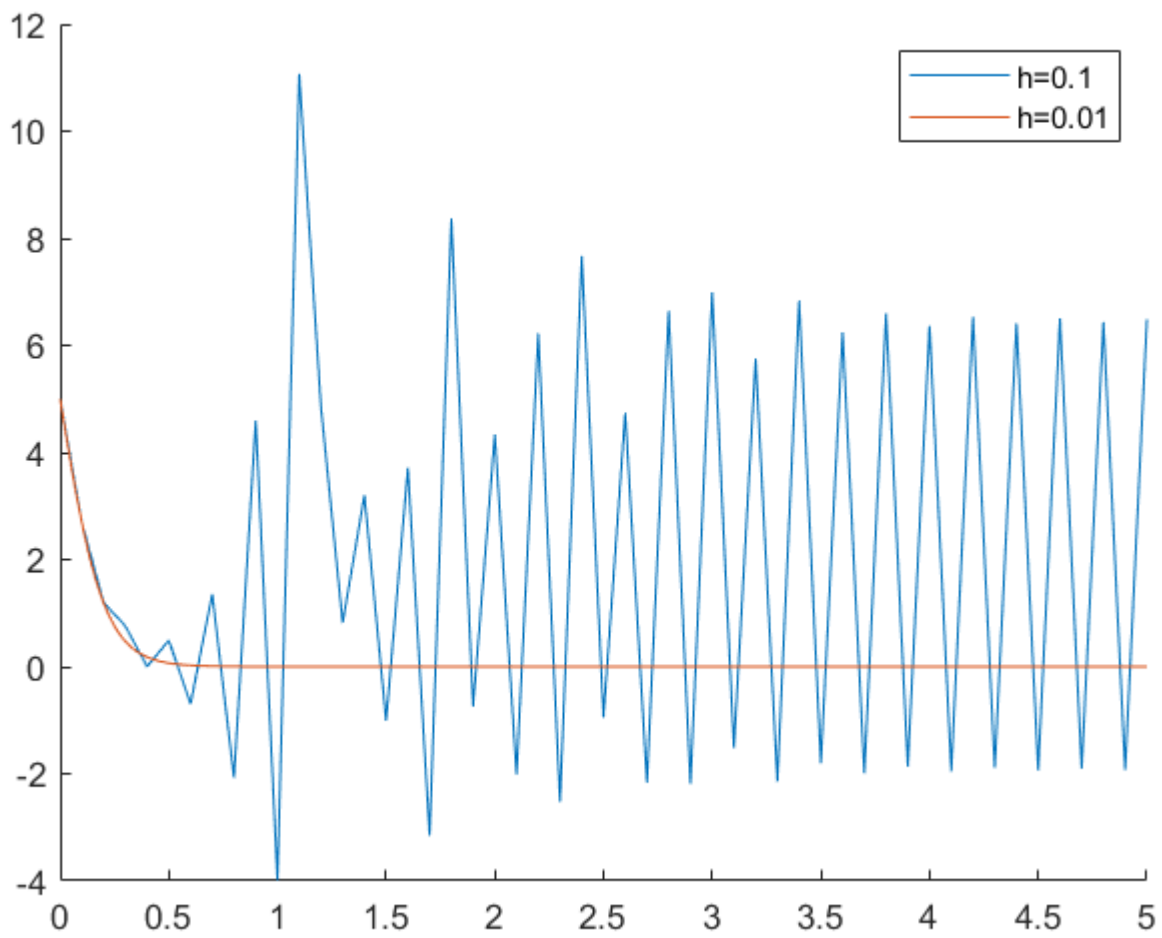


As seen here, this accurately describes what the function should look like, since this method and differential equation converge when $h=0.1$ and 0.01 . This is unlike the method in part b, whose radius of convergence does not contain those points (if it converges at all!).

Problem 2

```
function [result] = ABMethod3(f,y0,y1,y2,t0,tn,h)
    yn = [y0; y1; y2];
    i = t0;
    while i < tn
        v1 = f(i,yn(end,1));
        v2 = f(i-h,yn(end-1,1));
        v3 = f(i-2*h,yn(end-2,1));
        yn = [yn; yn(end,1) + h*((23/12)*v1 - (16/12)*v2 + (5/12)*v3)];
        i = i + h;
    end
    result = yn;
end
```

Output:



As seen here, the radius of convergence includes $h=0.01$ but not 0.1. Thus, for part a our solution does not reach the equilibrium as expected, but it does for part b when we lower h .